# Edbert Taidy

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## CAREER SUMMARY

**Computer Science graduate with Distinction, focused on full-stack software development. Experienced in building and deploying web and mobile applications using TypeScript, React, Vue.js, Flutter, Dart, Java, Spring Boot, Python, and SQL databases. Hands-on experience with REST APIs, authentication, CI/CD, and cloud deployment through independent projects and team-led development.**

## EDUCATION

**Singapore Institute of Management-University of Wollongong** Oct 2023 – Sept 2025

**Bachelor of Computer Science Majoring in Digital Systems Security and Cyber Security (Double Major), Singapore**

- Graduation Date: September 2025.
- Weighted Average Mark: 76.88 (Distinction).
- Achieved three high distinctions in **CSCI203 Algorithms and Data Structures, CSCI368 Network Security**, **CSIT328 Web Security**.

**Singapore Institute of Management (SIM), Singapore** Oct 2022 – Sept 2023

**Diploma in Information Technology**

## PROJECTS

**Full-Stack Developer | Meta Global AI Developer Hackathon** Oct 2026 – Nov 2026

**RoboRally — The Factory Rescue (Next.js, React, TypeScript, Three.js, FastAPI, Python, Pydantic, MongoDB, Muse Spark 1.3), Railway**

- Built an AI-agent-powered 3D maze game in **10 days** for the **Meta Global AI Developer Hackathon**, using **natural-language missions** to guide the robot.
- Integrated **Muse Spark 1.3** through the **Meta Model API** to produce constrained **JSON** plans in segments of up to 12 moves.
- Added a server-side rules engine that validates moves against walls, lasers, doors, triggers, enemies, and battery before step-by-step execution.
- Implemented stop-and-replan behavior for death, victory, and state changes, with completed levels and prompt history persisted to user accounts.
- Deployed the application on **Railway**.

**SignalBait Mobile — Android Message Scanner & Ticket Submission** Oct 2026 – Present

**Flutter, Dart, Java, Spring Boot, Python, FastAPI, PostgreSQL, Hugging Face Transformers**

- Developed an Android companion app for scanning suspicious messages and supporting user ticket submissions.
- Integrated the existing FastAPI scanning service and Spring Boot ticket API for text and image analysis, ticket submission, and ticket-status viewing.
- Connected the scanning workflow to a Hugging Face zero-shot classification model and added security guidance in the mobile interface.

**SignalBait Admin Dashboard — Administrative Ticket Management System** Sep 2026 – Present

**(Self-Directed Project)**

**Angular, TypeScript, Django, Python, PostgreSQL, ECharts, Tailwind CSS**

- Developed an administrative dashboard to **manage user reports, update ticket statuses, and monitor ticket activity**.
- Implemented **session-based authentication** to restrict dashboard access to administrators.
- **Built analytics visualizations using ECharts** and an audit-log interface to review administrative actions.
- **Integrated Angular frontend components with Django API endpoints and a shared PostgreSQL database** used by the main application.
- Deployed the frontend and backend as separate services using **Vercel and Render**.

**SignalBait — AI-Assisted Phishing & Scam Scanner** Sep 2026 – Present

**Vue.js, TypeScript, Tailwind CSS, Java, Spring Boot, Python, FastAPI, PostgreSQL, Hugging Face Transformers**

- Developed a full-stack web application that analyzes suspicious messages using an AI-assisted classification pipeline, supporting both text and image inputs.
- Integrated the third-party **facebook/bart-large-mnli** model through **Hugging Face Transformers** for **zero-shot text classification.**
- Built **REST APIs with Spring Boot and FastAPI** to connect the frontend with backend and machine learning services.
- Implemented ticket reporting and tracking workflows, **including email notifications for ticket creation and status updates** using **Brevo.**
- Deployed the frontend and backend services using **Vercel and Render**.

**Urgent or Is It? – Cybersecurity Awareness Game (Self-Directed Project)** Aug 2026 – Present

**React, TypeScript, Next.js, NestJS, MongoDB, Tailwind CSS, Vercel**

- **Developed and deployed** an independent full-stack cybersecurity awareness game using **React, TypeScript, NestJS, and MongoDB**, featuring interactive phishing-detection scenarios and persistent user progression.
- **Designed and implemented RESTful APIs** with NestJS and Mongoose for user registration, authentication, profiles, and **per-user game-state persistence**.
- **Implemented secure authentication and authorization** using **JWT, bcrypt password hashing, protected API endpoints, and DTO-based input validation** with NestJS guards and class-validator.
- **Built automated CI/CD workflows** with **GitHub Actions**, integrating production builds and deployment while configuring frontend-to-backend API communication across separate services.

**TreeVisual – Binary Tree Learning Tool (Self-Directed Project)** Feb 2026 – Present

**Python, PySide6**

- Designed and developed a desktop application to visualize **binary tree traversals** with step-by-step simulation.
- Implemented interactive controls to demonstrate **preorder, inorder, and postorder** traversal logic.
- Built save/load functionality for structured tree data using **JSON**.
- Focused on **clean UI design** to improve conceptual understanding of **data structures**.
- Packaged and distributed the application with downloadable builds for demonstration.

**Solo Participant | Nanyang Technological University x Base 3 Hackathon** Aug 2025 – Aug 2025

**Phish or Miss — A Phishing Awareness Game (React, Vite, Flask, AWS EC2)**

- Designed and built a **full-stack web application MVP within 5 days** in a competitive hackathon environment
- Deployed application on **AWS EC2 with Elastic IP**, enabling reliable public access for live demonstration during hackathon
- Implemented frontend-backend integration using REST APIs
- Debugged deployment and runtime issues under time pressure to ensure stable demo delivery

**Final Year Project Team Leader | SIM-UOW** Jan 2025 – May 2025

**AI-nteractive: Face Animated Learning Management System**

**(Flask, MongoDB, SadTalker API)**

- Led a team of 5 to integrate **AI-driven video generation** into a custom LMS
- Optimized processing pipeline, achieving **~90% reduction in generation time** using ffmpeg
- Designed backend workflows and API integration for automated content generation
- Applied **Agile/Scrum practices** to manage sprints and coordinate development

## VOLUNTEERING EXPERIENCE

**PPIS - Peduli Kartini Community Outreach Event**

**Logistics Assistant | Perhimpunan Pelajar Indonesia di Singapura (PPIS)**

May 2025

- Executed logistics support by distributing hampers and refreshments, and assisting participants in accessing legal and mental health services
- Collaborated with a team of 8 to coordinate setup and logistics for a 70+ participant community event

## CERTIFICATIONS

- **JLPT (Japanese Language Proficiency Test) N3 –** ***Awarded 27 Aug 2026***
- **AWS Certified Cloud Practitioner (CLF-C02) –** ***Awarded 7 Aug 2025***
- **ISC2 Certified in Cybersecurity (CC) –** ***Awarded 29 Jul 2025***

## SKILLS

- **Languages: Python, Java, JavaScript, TypeScript, Dart, C++, PL/SQL**
- **Frontend: React, Next.js, Vite, Flutter, HTML, CSS, Tailwind, Angular, Vue**
- **Backend:** NestJS, Flask, PHP, REST APIs, Django, FastAPI, Spring Boot
- **Databases:** MongoDB, MySQL, PostgreSQL
- **Cloud & DevOps:** AWS (EC2, S3, CloudFront, Elastic IP), Vercel, Railway, Git, GitHub Actions, CI/CD, HTTPS
- **Practices:** Data Structures & Algorithms, OOP, Secure Coding, OWASP Top 10, Agile/Scrum, Debugging & Testing